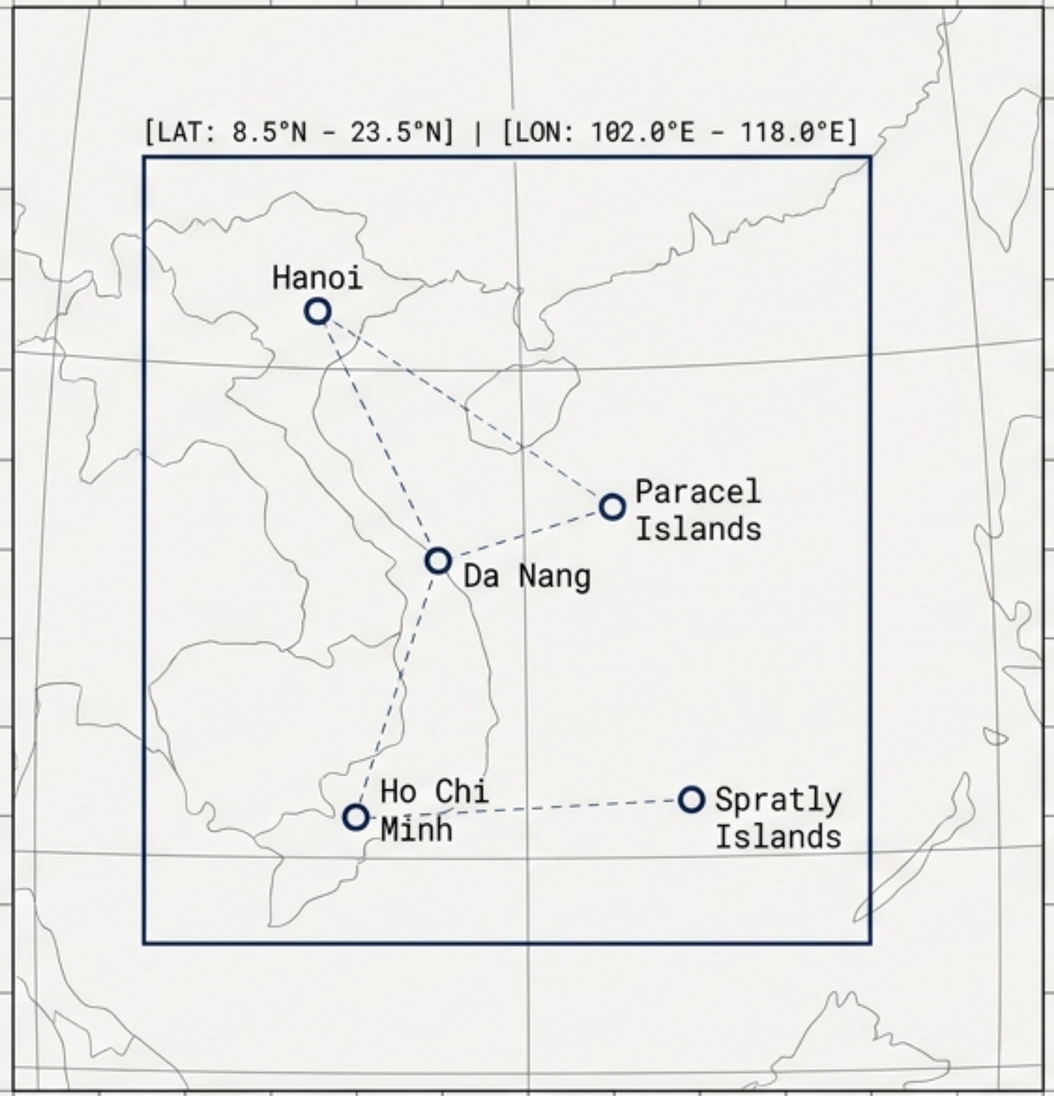
SYS_STATUS: VERIFIED
SYS_MARGINS: NON-NEGATIVE
DOC_REF: FOC-CAPSTONE-FINAL

Vietnamese Earth Observation Constellation

Capstone System Architecture & Verification Report

Gemini Notebook



Mission Context & Operational Scope

Mission Statement

Provide continuous, high-revisit Earth observation coverage over the entire landmass and maritime zones of Vietnam for a 5-year primary operational lifecycle.

Primary Capability

Disaster monitoring, maritime domain awareness, and environmental management via high-resolution optical imagery.

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System Requirements Flowdown

SYS-REQ-01 [REVISIT]

Mean regional revisit time.

< 30.0 min

SYS-REQ-02 [ORBIT]

Sun-Synchronous Orbit (SSO) with ±5 km deadband.

500 - 600 km

SYS-REQ-03 [PROPULSION]

Minimum total Delta-v capacity.

≥ 300.0 m/s

SYS-REQ-04 [DISPOSAL]

EOL atmospheric re-entry duration.

< 2.0 years

SYS-REQ-05 [PAYLOAD]

Active off-nadir cross-track steering.

≥ ± 30.0°

SYS-REQ-06 [MASS]

Maximum single-satellite wet mass.

≤ 150.0 kg

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Trade Space & Optimization Architecture

Design Variables (x)

N (Total Satellites):
[8, 10, 12, 14, 15, 16, 18, 20, 24]

P (Orbital Planes):
[2, 3, 4, 6] (where S = N/P ≥ 2)

F (Phasing Parameter):
[0, 1, …, P-1]

h (Altitude): [500 km, 600 km]
SSO (frozen eccentricity e=0)

Cost Function Logic (J(x))

Minimize N
(Hardware Procurement Penalty)

↓

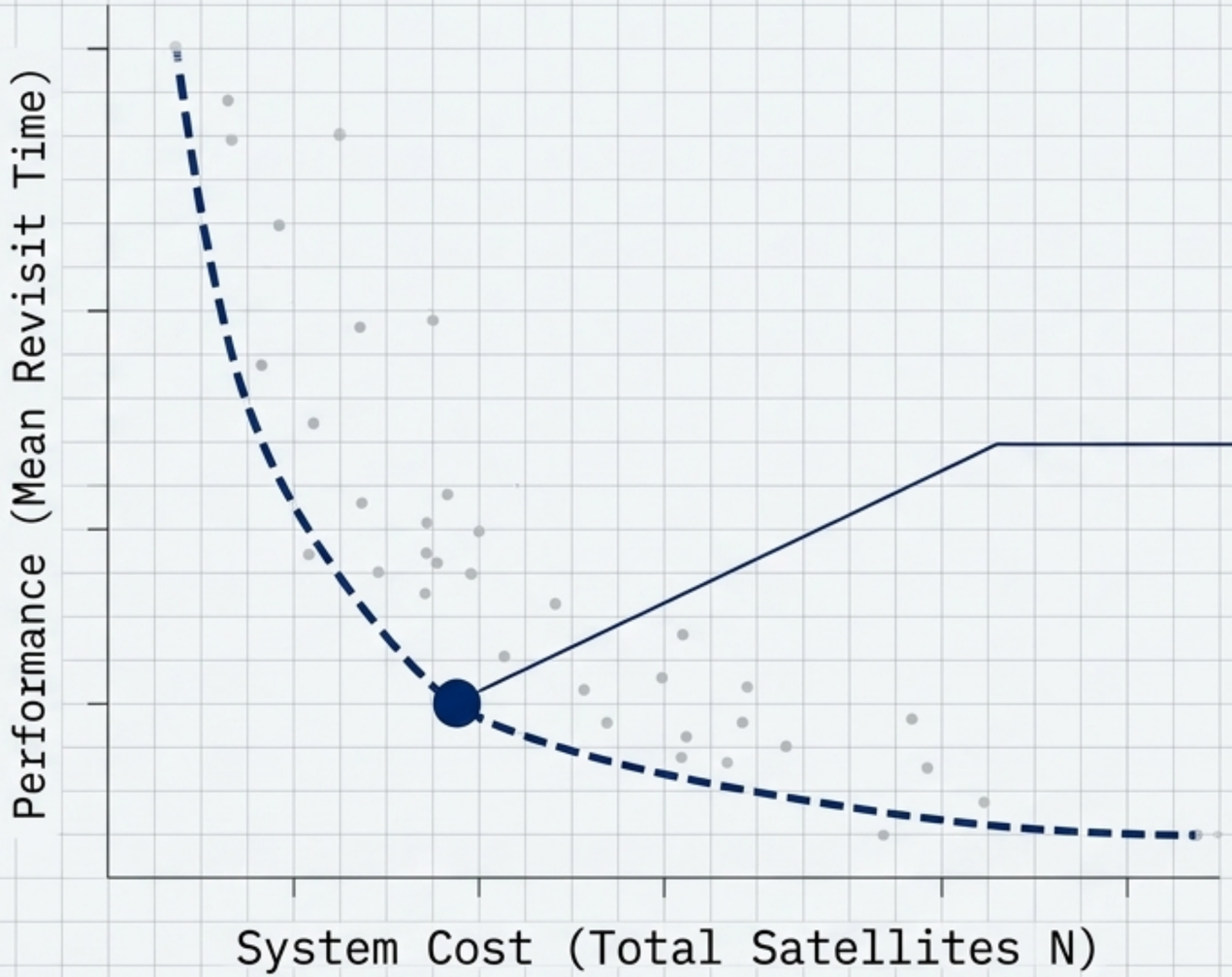
Minimize P
(Launch Vehicle / Plane Deployment Penalty)

↓

Subject to inequalities:
$g(x) \leq 0$ (SYS-REQ compliance)

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Exhaustive Global Grid Search



Search Scope: 79 unique candidate constellation architectures evaluated.

Methodology: Brute-force optimization across discrete, non-convex Walker Delta parameters.

The Optimum: The mathematical winner that balances acquisition lifecycle costs against strict SYS-REQ-01 compliance.

Optimization is finding the best design, not just a working one.

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Optimal Baseline Architecture

Walker Delta 97.4°: 18/3/1

System Dimensions	Operational Characteristics
Total Fleet (N): 18 Satellites Orbital Planes (P): 3 Planes Satellites per Plane (S): 6 SVs/Plane Inter-plane Phasing (F): 1	Altitude: 500.0 km SSO Chosen for superior Ground Sample Distance and 1563.60 km footprint radius. Agility Integration: ±30° active cross-track roll steering.

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High-Fidelity 48-Hour Numerical Simulation

Environmental Inputs

- Payload Variables: Active ±30.0° cross-track roll steering. steering.
- Distortion Filter: 10.0° elevation mask applied.

Propagator Engine

- Engine: Custom Python-based orbital propagator.
- Temporal Resolution: 48.0 Hour Horizon
- Time Steps: 30.0s (5761 Total Steps)
- Note: Captures complete Earth rotation & repeat cycles.

Geographic Grid

- Spatial Resolution: 132 Target Grid Points
- Step Size: 1.5° × 1.5° steps over bounding box.
- Strategic Targets: 5 critical nodes actively logged.

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Simulation Results: Revisit Dynamics & Payload Steering

Passive Geometry

Single-plane raw revisit gap without agility integration is:

15.77 min

Active Operations

By integrating 3 planes with active ±30.0° cross-track roll steering (SYS-REQ-05), the regional mean revisit time drops to:

25.70 minutes

[SYS-REQ-01 COMPLIANT]

(Clears the 30.0 min strict threshold).

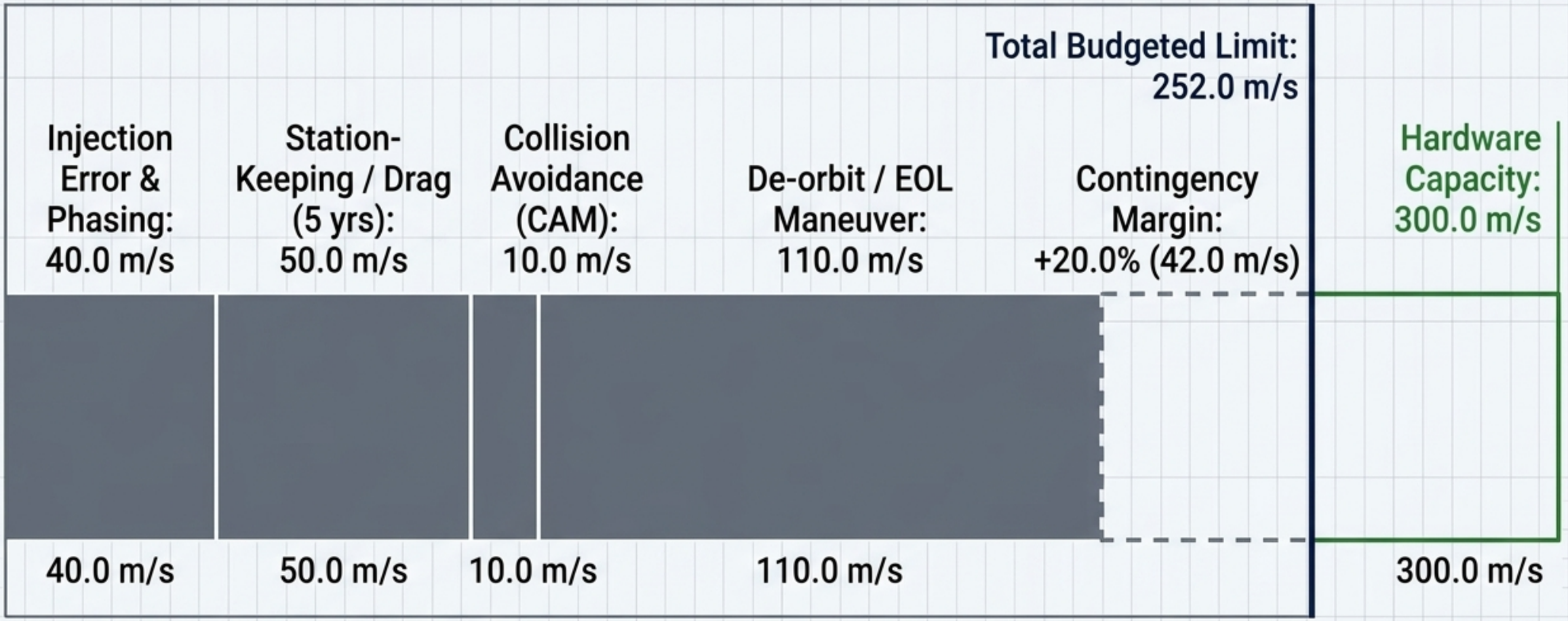
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Master Verification Matrix (Non-Negative Margins)

REQ-01 (Revisit) Required <30.0m | Achieved 25.70m | Margin +14.33%		PASS
REQ-02 (Orbit) Required 500km ±5km | Achieved 500km J2 Propagated | Margin 0.0% (Nominal)		PASS
REQ-03 (Delta-v) Budgeted 252m/s | Hardware 300m/s | Margin +19.05%		PASS
REQ-04 (Disposal) Required <2.0 yrs | Achieved Re-entry < 2.0 yrs | Margin +20.0%		PASS
REQ-05 (Steering) Required ±30.0° | Achieved ±30.0° Integrated | Margin 0.0%		PASS
REQ-06 (Mass) Required ≤150.0kg | Achieved 150.0kg Allocated | Margin 0.0% (Boundary)		PASS

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Technical Budgets: Lifetime Delta-v Allocation

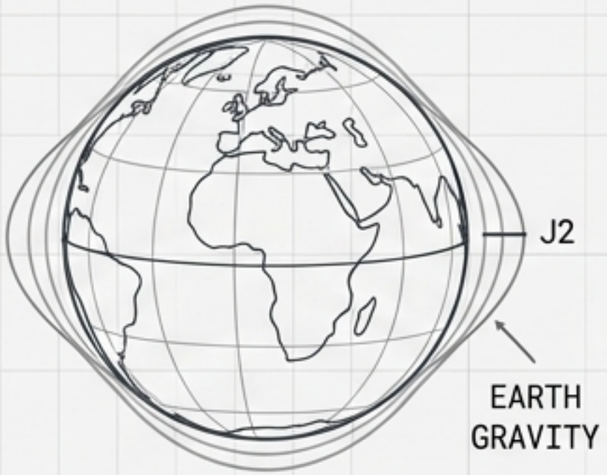


Provides a definitive **+19.05% margin** against physical hardware limits, ensuring SYS-REQ-03 and SYS-REQ-04 robustness over the 5-year lifecycle.

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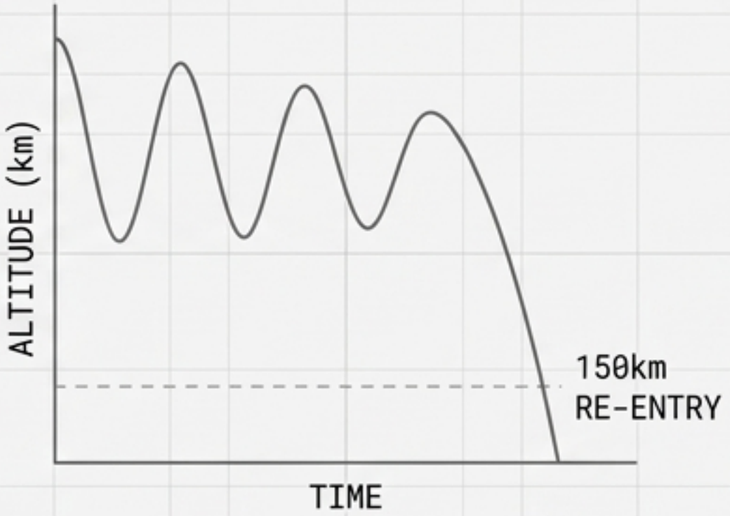
Systemic Assumptions & Dynamical Realities

Orbital Perturbations



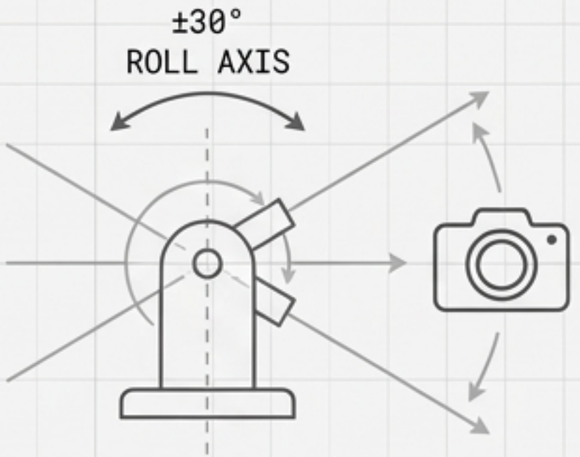
High-fidelity verification assumes non-spherical Earth gravity (J2 perturbation model) over the 500km altitude timeline.

Atmospheric Drag



EOL decay calculations (SYS-REQ-04) utilize the Jacchia 1970 density model to confirm rapid re-entry from the 150km lowered lowered perigee.

Hardware Reliability



The 25.70 minute revisit time is strictly contingent on the mechanical reliability of the ±30° payload roll steering mechanisms over 5 years.

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Verdict & FOC Validation

The Walker Delta 18/3/1 architecture at 500 km SSO represents a fully converged engineering design.

Through exhaustive grid search optimization and 48-hour high-fidelity propagation, all constraints are satisfied.

System validated for Full Operational Capability (FOC) deployment with strict non-negative margins across all mission parameters.

VERIFICATION CYCLE: COMPLETE

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